

① Find the particular solution of the differential equation $2ye^{\frac{x}{y}} dx + (y - 2xe^{\frac{x}{y}}) dy = 0$ given that $x=0$ when $y=1$.

② Find $\int \frac{x^2 + x + 1}{(x^2 + 1)(x + 2)} dx$

③ Find the values of p and q for which

$$f(x) = \begin{cases} \frac{1 - \sin^3 x}{3\cos^2 x} & ; \text{ if } x < \pi/2 \\ p & ; \text{ if } x = \pi/2 \\ \frac{q(1 - \sin x)}{(\pi - 2x)^2} & ; \text{ if } x > \pi/2 \end{cases}$$

is continuous at $x = \pi/2$.

④ Prove that:

$$\tan^{-1}\left(\frac{6x - 8x^3}{1 - 12x^2}\right) - \tan^{-1}\left(\frac{4x}{1 - 4x^2}\right) = \tan^{-1} 2x ;$$

for $|2x| < \frac{1}{\sqrt{3}}$

⑤ Find the acute angle between the plane $5x - 4y + 7z - 13 = 0$ and the y -axis.

⑥ Find the shortest distance between the following lines:

$$\frac{x-3}{11} = \frac{y-5}{-2} = \frac{z-7}{1} \quad \text{and} \quad \frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$$

⑦ Give a simple chemical test to distinguish between the following pair

(i) $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{COCH}_3$

(ii) $(\text{CH}_3)_2\text{NH}$ and $(\text{CH}_3)_3\text{N}$

(iii) Benzaldehyde and Benzoic acid.

⑧ Which halogen compound in each of the following pairs will react faster in $\text{S}_\text{N}2$ reaction:

(i) CH_3-Br or CH_3-I

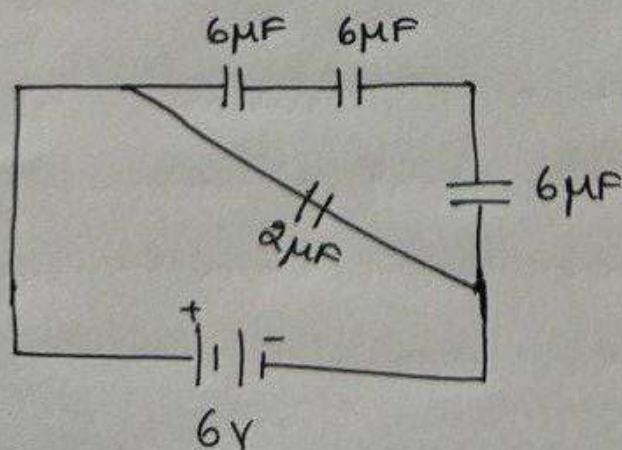
(ii) $(\text{CH}_3)_3\text{C}-\text{Cl}$ or CH_3-Cl

⑨ Calculate the freezing point of the solution when 31g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) is dissolved in 500g of water. K_f for water is 1.86 K kg/mol

⑩ The rate constant of a first order reaction increases from 2×10^{-2} to 4×10^{-2} when the temperature changes from 300K to 310K. Calculate the energy of activation (E_a).

$$(\log 2 = 0.3010, \log 3 = 0.4771, \log 4 = 0.6021)$$

- ⑪ Differentiate between a dielectric and a conductor?
- ⑫ why alloys like constantan and manganin are used for making resistors?
- ⑬ Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity. Explain, give reason.
- ⑭ An electric dipole of length 4cm , when placed with its axis making an angle of 60° with the uniform electric field, experiences a torque of $4\sqrt{3}\text{Nm}$. Calculate the potential energy of the dipole, if it has charge $\pm 8\text{nC}$.
- ⑮ Four capacitors of values $6\mu\text{F}$, $6\mu\text{F}$, $6\mu\text{F}$ and $2\mu\text{F}$ are connected to a 6V battery as shown in the figure.



- (i) Equivalent capacitance of the network
- (ii) charge on each capacitor.